

KATHI CHETAN REDDY

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B SECTION

LINEAR SEARCH

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[] = {10, 20, 30, 40, 50};
```

```
    int key = 30;
```

```
    int i;
```

```
    for (i = 0; i < 5; i++) {
```

```
        if (a[i] == key) {
```

```
        printf("element found at index %d", i);  
        return 0;  
    }  
}
```

```
printf("element not found");  
return 0;  
}
```

OUTPUT

element found at index 2

BINARY SEARCH

```
#include <stdio.h>  
  
int main() {  
    int a[] = {10, 20, 30, 40, 50};  
    int key = 40;  
    int l = 0, r = 4, m;
```

```
while (l <= r) {  
    m = (l + r) / 2;  
  
    if (a[m] == key) {  
        printf("element found at index %d", m);  
        return 0;  
    }  
    else if (key < a[m]) {  
        r = m - 1;  
    }  
    else {  
        l = m + 1;  
    }  
}  
  
printf("element not found");  
return 0;  
}
```

OUTPUT

element found at index 3

The screenshot displays the Programiz Online C Compiler interface. The browser address bar shows the URL `programiz.com/c-programming/online-compiler/`. The Programiz logo and "C Online Compiler" text are at the top left, with a "Programiz PRO" button at the top right. The main workspace is divided into a code editor on the left and an output window on the right.

The code editor contains the following C code:

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {10, 20, 30, 40, 50};
5     int key = 30;
6     int i;
7
8     for (i = 0; i < 5; i++) {
9         if (a[i] == key) {
10             printf("element found at index %d", i);
11             return 0;
12         }
13     }
14
15     printf("element not found");
16     return 0;
17 }
18
```

The output window displays the result of the program execution:

```
element found at index 2
=== Code Execution Successful ===
```

The Windows taskbar at the bottom shows the system clock as 7:08 PM on 8/7/2026.

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C Online Compiler

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main.c

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Run

1#include <stdio.h>

2int main() {

3int a[] = {10, 20, 30, 40, 50};

4int key = 40;

5int l = 0, r = 4, m;

6

7while (l <= r) {

8m = (l + r) / 2;

9

10if (a[m] == key) {

11printf("element found at index %d", m);

12return 0;

13}

14else if (key < a[m]) {

15r = m - 1;

16}

17else {

18l = m + 1;

19}

20}

21

22printf("element not found");

Output

Clear

element found at index 3

=== Code Execution Successful ===

JS

TS

Type here to search



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ENG IN

7:09 PM

8/7/2026